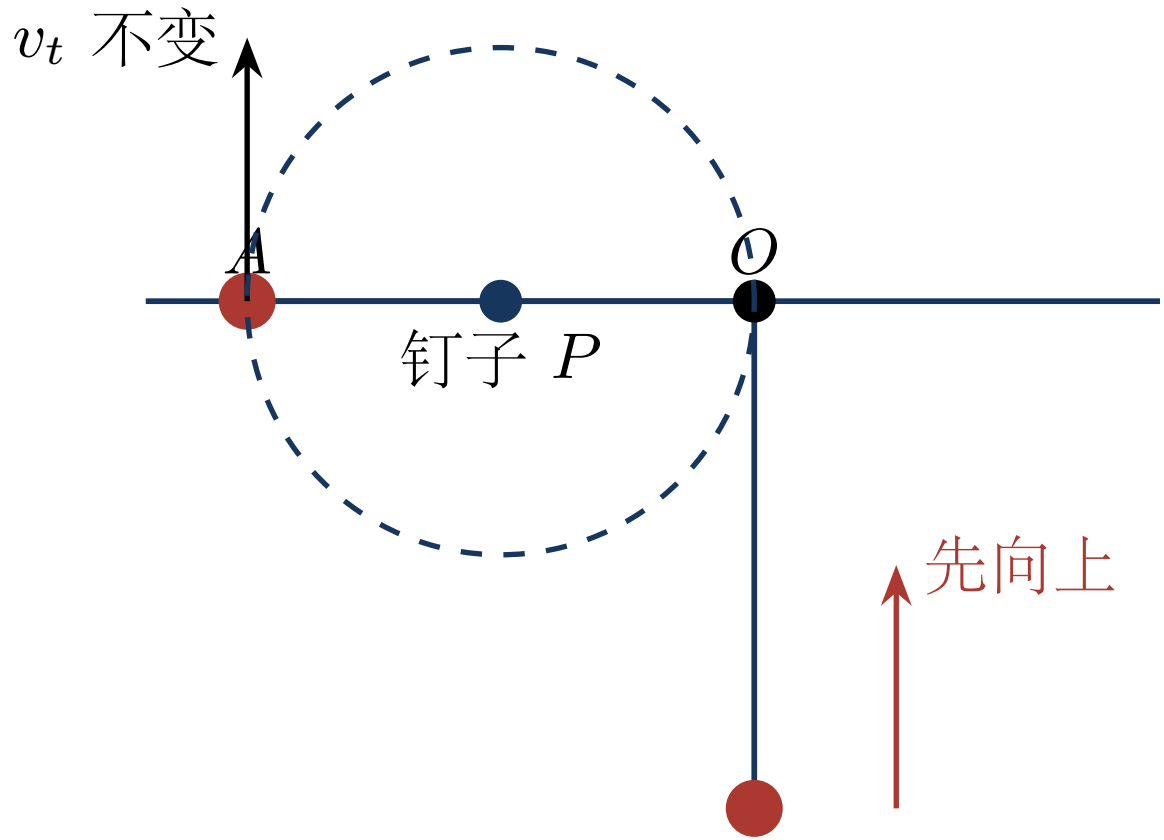

新半径 $R_0 = l_0/2$，新角动量 $J_P = mv_tR_0$